

## Problem

Given an array of n integers  $x_1, x_2, \dots, x_n$ , our task is to find the maximum sub array sum, i.e., For example, in the array  $[-1, 2, 4, -3, 5, 2, -5, 2]$  the following subarray produces the maximum sum 10:  $[-1, 2, 4, -3, 5, 2, -5, 2]$

## Input Format

Two line input First line should read an integer value which represents size of array Second line consists list of elements(integers) separated by space

## Constraints

1

## Output Format

It prints integer value which represents maximum sub array sum

## Sample Input 0

```
8
-1 2 4 -3 5 2 -5 2
```

## Sample Output 0

10

Change Theme

Language

Java 7



```

1  import java.io.*;
2  import java.util.Scanner;
3  class Solution{
4      public static void main(String [] args){
5          Scanner sc = new Scanner(System.in);
6          if(!sc.hasNextInt()){
7              return;
8          }
9          int n = sc.nextInt();
10         int [] arr = new int[n];
11         for(int i=0; i<n; i++){
12             arr[i] = sc.nextInt();
13         }
14         int max = arr[0];
15         int sum = arr[0];
16         for(int i=1; i<n; i++){
17             sum = Math.max(arr[i],sum+arr[i]);
18             max = Math.max(sum,max);
19         }
20         System.out.println(max);
21     }
22 }
```